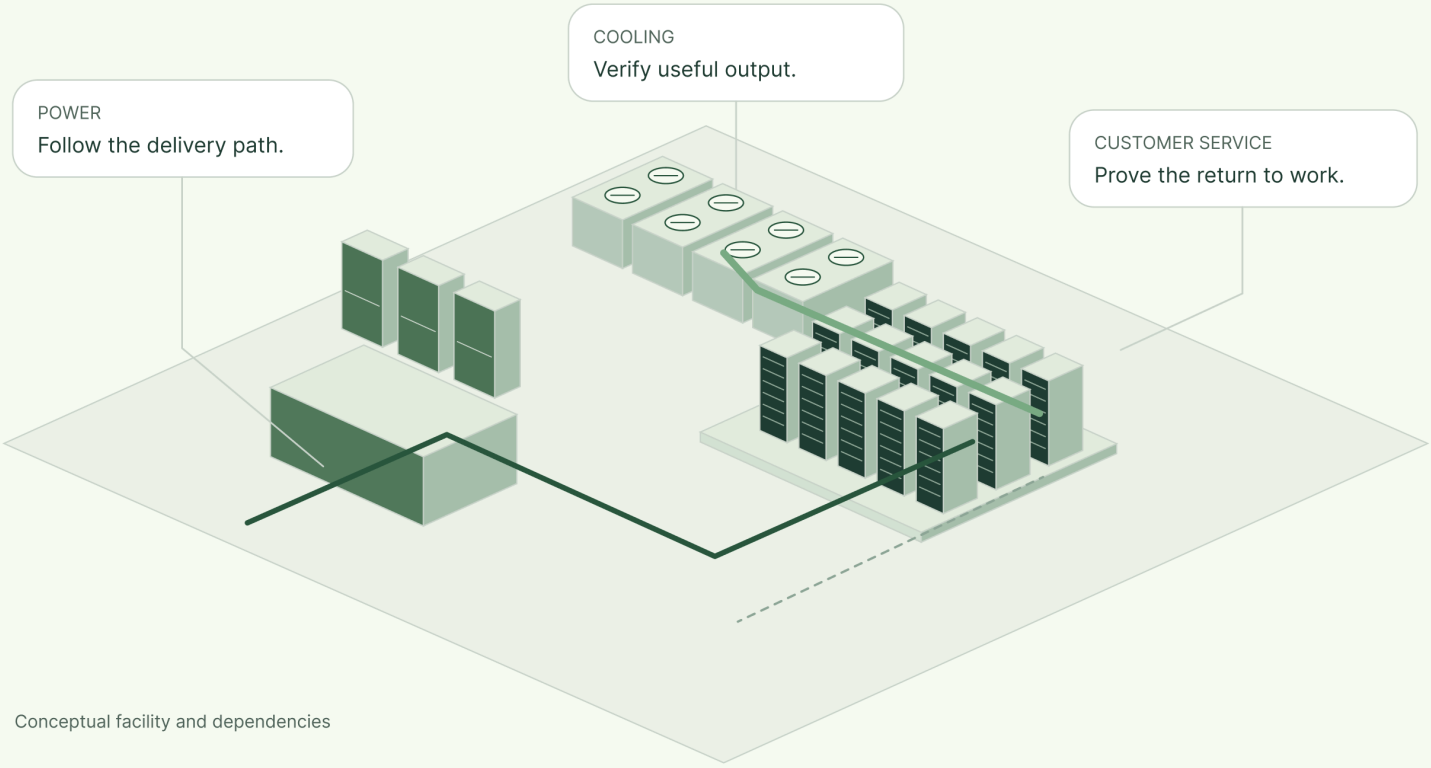
entangl

2025 / 2026

How data center outages escalate

The failures. The recovery. The decisions that shape both.



Conceptual facility and dependencies

AN ILLUSTRATED FIELD GUIDE

Lessons from 12 months of U.S. incidents and practical ways to put Entangl to work.

14 SEPTEMBER 2025 - 14 SEPTEMBER 2026

52 PAGES

THE ENTANGL PERSPECTIVE

Make the next decision with the right evidence.

An approved plan. An alarm. A recovery call.

The hard part is keeping the equipment, the evidence and the decision connected as the situation changes. That is the operating problem this report explores, and the work Entangl is built to support.

WHAT CHANGES WHEN THE NEXT DEPENDENCY FAILS?

The work boundary.
The evidence available.
The person who must decide.

Make the next decision with the site in view.

Pick a familiar procedure or incident. Bring the evidence. Judge Entangl against work your engineers already understand.

Explore Entangl

WHAT THE CASES REVEAL

Five gaps worth looking for at your site.

A recurring opportunity: connect engineering knowledge to the decision being made.

01 Scope

Cases 04-05

The difference between the intended task and every asset the issued package can affect.

Nebius, March / p. 18

02 Sequence

Case 06

The prerequisites and release conditions carried into the technician's instructions.

Google, Iowa / p. 24

03 State

Cases 02-03

The difference between the documented arrangement and the equipment's observed condition.

Azure, May / p. 15

04 Context

Case 01

Whether available signals lead to an accountable investigation with the right references.

Nebius, August / p. 9

05 Recovery

Cases 02, 08-09

The access, dependencies and customer tests required after physical infrastructure returns.

Azure, February / p. 12

Use these themes to choose a review task. They describe this selected evidence base, not the frequency of failures across the industry.

Entangl analysis of the case chapters, pages 9-35. Source register: pages 49-51.

INSIDE THE REPORT

Read the failure. Follow the recovery.

Each detailed case has three parts: what happened, the recovery story, and a practical Entangl application.

01	Nebius	Cooling, detection and restart	09
02	Microsoft Azure	Transformer and transfer failure	12
03	Microsoft Azure	Cooling protection and manual recovery	15
04	Nebius	Maintenance beyond its scope	18
05	Microsoft Azure	The combined optical-change impact	21
06	Google Cloud	Work-order sequencing in Iowa	24
07	Google Cloud	Reduced network capacity in Oregon	27
08	AWS / Coinbase	The customer recovery boundary	30
09	Blacksmith	Storage, access and degraded operation	33

The operating argument

Perspective, evidence and recurring questions. Pages 2-8.

Broader incident record

Additional reports and verification leads. Pages 36-41.

Put it to work

Workshops, evaluation and sources. Pages 42-51.

SCOPE AND EVIDENCE

A public evidence base. A practical operating lens.

12
MONTHS

9
DETAILED CASES

32
REPORTS + LEADS

The period

September 14, 2025 through September 14, 2026. Selection follows incident dates.

The scope

U.S. power, cooling, shared infrastructure and local or inter-campus network disruption.

Operator post-mortems, status histories, company statements, news and archived updates form the evidence base. The source register retains uncertainty and possible overlap.

Company RCA	The affected provider explains the mechanism.
Customer RCA	A downstream account with its own perspective.
Statement / status	Useful impact detail; a full cause may be absent.
Archive / lead	Evidence that still needs further verification.

This is not a certified count of distinct facility outages. Private incidents, customer-only RCAs, inaccessible notices and possible duplicates limit completeness. Naming a company does not imply endorsement.

Incident mechanisms are simplified in the diagrams. Geometry and equipment counts are illustrative unless explicitly stated. Entangl applications are proposed evaluations; they are not claims of involvement or prevention.

THE RECOVERY GAP

The facility returned. The service was still recovering.

Two Azure incidents show why restoration needs more than one milestone.



Physical restoration

Dark green: power or cooling restored.

Continuing recovery

Pale green: remaining time until final service-impact mitigation.

Calculated from the published UTC milestones. These are different facility milestones, and individual customer impact varied. The timelines in the case chapters show the service recovery stages.

THE OPERATING LESSON

Define what restored means at every layer.

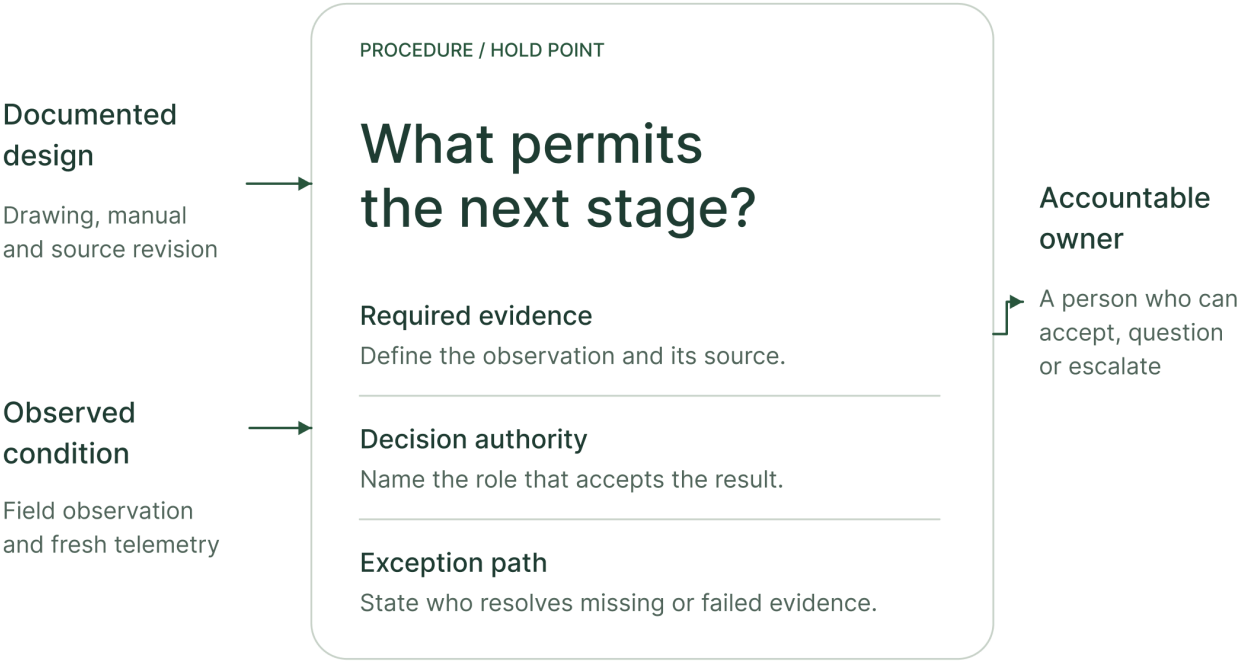
Use Entangl to review the recovery procedure with its site sources, prerequisites and service-owner handoffs.

R13: RCA / Video / DCD | R19: RCA / Video / Essay

AN OPERATING MODEL

Connect the evidence to the next decision.

ANATOMY OF A REVIEWABLE DECISION / ILLUSTRATIVE EXAMPLE



The source, the current condition and the authority establish different things.

These sources become useful through their relationship. A reviewer can follow a step to its evidence; a responder can connect an alarm to equipment and a relevant operating reference.

ENTANGL APPLICATION

A source gives the decision a foundation.
A named owner makes it actionable.

Entangl site model / Entangl telemetry | Illustrative annotation, not an operating instruction.

NINE DETAILED CASES

Find your operating pressure.

Start with the question that feels closest to the work at your site.

01

Nebius / Aug

Monitoring independence

READ CASE / P. 09

02

Azure / Feb

The full supply path

READ CASE / P. 12

03

Azure / May

Equipment protection state

READ CASE / P. 15

04

Nebius / Mar

Changed maintenance scope

READ CASE / P. 18

05

Azure / Jul

Combined change impact

READ CASE / P. 21

06

Google / Iowa

Instructions in the field

READ CASE / P. 24

07

Google / Oregon

Remaining capacity

READ CASE / P. 27

08

AWS / Coinbase

Failure-domain exposure

READ CASE / P. 30

09

Blacksmith

Recovery dependencies

READ CASE / P. 33

Every case shows the failure mechanism, follows the recovery and turns the lesson into a useful operating review.

Company and customer post-mortems: pages 9-35. Linked evidence: pages 49-51.

CASE 01 / NEBIUS

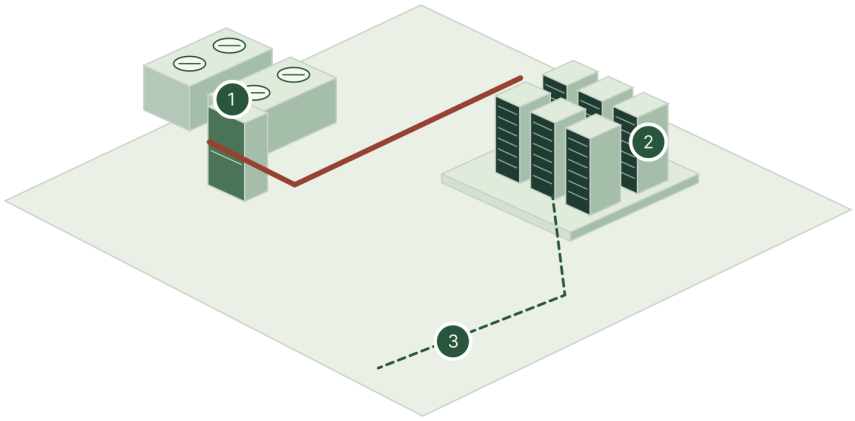
The cooling failed. The recovery exposed more.

August 19-20, 2026 | us-central1, United States

A storm disabled building management and the chilled-water cooling loop. Heat shut down shared infrastructure. About one third of running virtual machines were interrupted. Stale disk attachments, a disk hot-plug bug and constrained restart automation extended the recovery. Nebius reported the disk bug fixed and committed to independent environmental alerts and regional recovery drills.

Cooling, heat and resource state

The physical event creates several recovery tasks.



01 **Storm exposure**

Building management and the chilled-water loop failed.

02 **Thermal impact**

Heat shut down shared infrastructure.

03 **Resource recovery**

Disk state and restart automation extended the work.

Cooling restored was a milestone. Resource recovery continued.

Published mechanism with conceptual architectural artwork; site geometry and equipment counts are illustrative.

6 hours

Approximate time to stabilize the facility thermally

A local cooling failure spread into shared infrastructure.

Thermal recovery and resource recovery were different jobs.

R28: RCA

CASE 01 / THE RECOVERY STORY

Make the response survive the failure it observes.

PUBLISHED MILESTONES

- **09:15 UTC**
  Rising temperatures visible; no alert.
- **10:15**
  Component-temperature alerts page engineers.
- **12:50**
  External VM connectivity restored.
- **21:58**
  Managed Kubernetes clusters healthy.
- **Aug 20, 08:38**
  Reserved GPU capacity restored.

August 19 unless stated. On-demand node repairs continued for days.

Sequence shown; spacing is not elapsed time.

01 **A signal needs a decision owner.**

A temperature trend becomes operationally useful when someone owns its interpretation and knows which response to initiate. Map the first credible warning to an accountable person, an escalation route and an approved procedure. Review the handoff as carefully as the sensor itself.

02 **Distinguish visibility from independence.**

Two dashboards can share a collector, a power source or an authentication service. Draw the monitoring path all the way to the responder. Ask what remains observable when local management, networking or the usual control room view becomes unavailable.

03 **Practise the return of customer resources.**

A rehearsal should end when the intended workload behaves correctly. Include interrupted resources, inconsistent attachment state and failed first attempts. Give each unresolved item an owner so that the last portion of recovery does not become an unstructured queue.

TAKE THIS QUESTION TO YOUR TEAM

Would your response still work if the system reporting the failure failed too?

R28: RCA | Operating lessons: Entangl analysis.

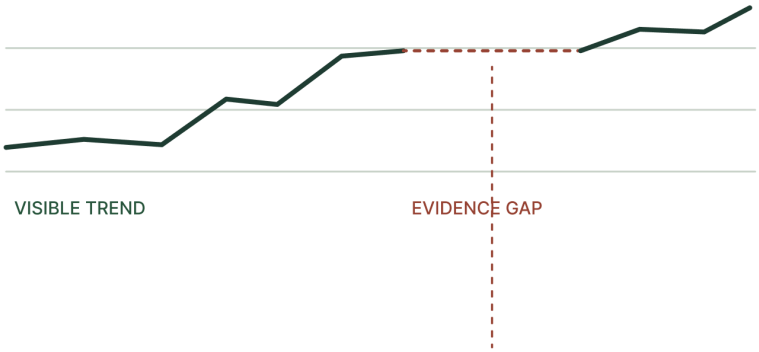
CASE 01 / WHERE ENTANGL HELPS

Turn a cooling alarm into a coordinated response.

Entangl can help your team connect available telemetry to site information and the procedure that supports the next decision. Use this scenario to evaluate the journey from the first credible signal to an assigned investigation.

ENTANGL EVALUATION / CONCEPTUAL REVIEW CANVAS

Can the warning still reach its owner?



Check the whole monitoring path.

Point, collector, network, responder

Which dependency fails with the plant?

Record source, freshness, missing evidence and the next verification owner.

Illustrative signal trace; no measured data or automatic detection is asserted.

Bring: site context: Cooling assets, operating sequences and affected loads; incident context: Timestamped trends, alarm history and known blind spots; response context: Approved EOPs, contacts and escalation responsibilities.

Investigate with context

Relate the available signals to the equipment and relevant source documents. Ask the operator to explain the evidence for the suspected condition and what remains uncertain.

Prepare the handoff

Review the cooling-loss procedure for the person responsible, the information needed from the facility operator and the conditions that require escalation. Rehearse a loss of the normal monitoring view.

Measure a useful result

Compare time to an assigned investigation, missing context and handoff completeness against the existing workflow. Test stale data as deliberately as a clear alarm.

Implementation focus. Independent sensors, resilient communications and cooling-system design require engineering outside this workflow. Telemetry connections and response actions must be scoped for the deployment.

Explore this with Entangl →

Explore the Entangl workflow | Conceptual evaluation; not a product screenshot.

CASE 02 / MICROSOFT AZURE

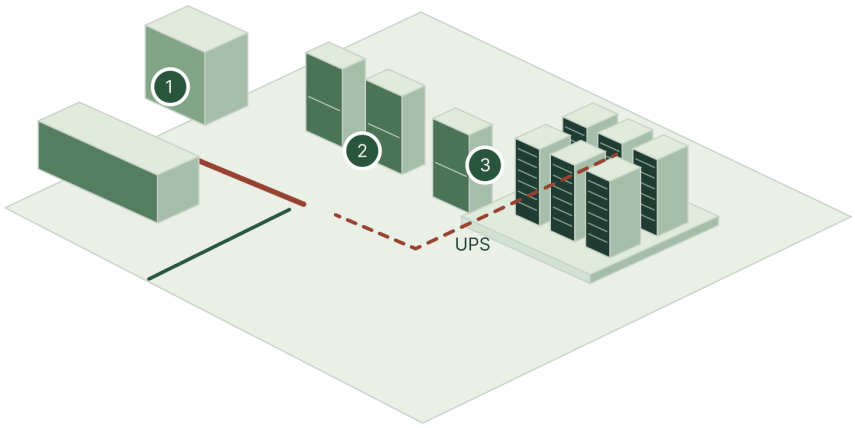
The generators started. The load did not transfer.

February 7-8, 2026 | West US, California

An onsite transformer failed while the utility grid remained available. Generators started, but a control-system failure prevented load transfer; UPS batteries depleted. Storage boot failures and recovery congestion extended the incident after power returned. Microsoft reported electrical-control changes and work on recovery sequencing and prioritization.

Available source. Broken transfer.

Trace source, control and stored energy to the load.



01 Source failure

The onsite transformer failed; the utility grid remained available.

02 Transfer failure

Generators started. The control system did not transfer the load.

03 Stored energy ends

UPS depletion interrupted power to dependent systems.

Available generation still needs a complete path to the load.

Published mechanism with conceptual architectural artwork; site geometry and equipment counts are illustrative.

20h 26m

Published window through final service-impact mitigation

Starting the generators did not complete the supply path.

Transfer logic sat between available power and the load.

R13: RCA / Video / DCD

CASE 02 / THE RECOVERY STORY

Follow the path all the way to service.

PUBLISHED MILESTONES

- **Feb 7, 07:58 UTC**
  Customer impact begins.
- **11:29**
  All facility power restored on generators.
- **19:05**
  Storage recovery complete.
- **23:30**
  Compute recovery complete.
- **Feb 8, 04:24**
  All affected services mitigated.

Different services and customers had different impact windows.

Sequence shown; spacing is not elapsed time.

01 **An available component is not an available path.**

A generator status does not by itself prove delivery to every required load. A useful operating review identifies the source, transfer logic, intermediate equipment and endpoint. The evidence for each part may come from different people or systems.

02 **Restore prerequisites before dependent work.**

Treat restart order as an engineering decision. For each service, identify what must already be healthy, what can be recovered independently and who confirms readiness. Otherwise teams can spend effort restarting components that are still waiting on a prerequisite.

03 **Design the recovery queue for scale.**

A procedure for one failed host does not describe the coordination burden of a broad restart. A rehearsal should include a surge of work, competing priorities and a clear way to distinguish a retryable condition from a problem requiring manual intervention.

TAKE THIS QUESTION TO YOUR TEAM

Does your recovery plan prove that power reached the load and that the workload came back?

R13: RCA | Operating lessons: Entangl analysis.

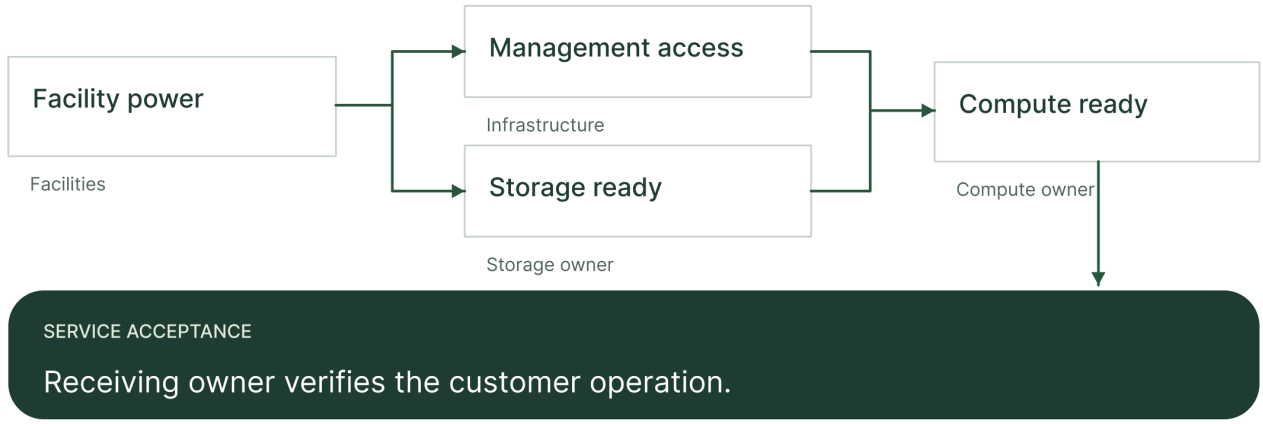
CASE 02 / WHERE ENTANGL HELPS

Review recovery against the actual site.

Entangl brings one-lines, equipment records and operating documentation into procedure preparation and review. Apply that context to a recovery plan so reviewers can challenge its prerequisites and handoffs before an incident.

ENTANGL EVALUATION / CONCEPTUAL REVIEW CANVAS

Recovery is a dependency graph.



Review the site-specific graph and evidence for every handoff; relationships shown are illustrative.

Bring: electrical context: Current one-lines, equipment identities and transfer sequence; recovery plan: Existing EOP, restart responsibilities and required evidence; acceptance criteria: The specific conditions each receiving team must verify.

Trace the documented relationships

Use the site model to inspect the named equipment and supporting sources. Require current operating-state evidence wherever a decision depends on a live condition.

Review the handoff conditions

Assess the recovery procedure for missing prerequisites, ambiguous responsibilities and premature completion claims. Give the engineer a source and a decision to resolve for each finding.

Rehearse the sequence

Walk the plan with facilities and service owners. Measure unresolved dependencies and time to an accepted recovery plan, including the effort required to correct the draft.

Implementation focus. Entangl does not replace transfer-system engineering or repair cloud storage software. The proposed application supports the operating plan and its review.

Explore this with Entangl →

Explore the Entangl workflow | Conceptual evaluation; not a product screenshot.

CASE 03 / MICROSOFT AZURE

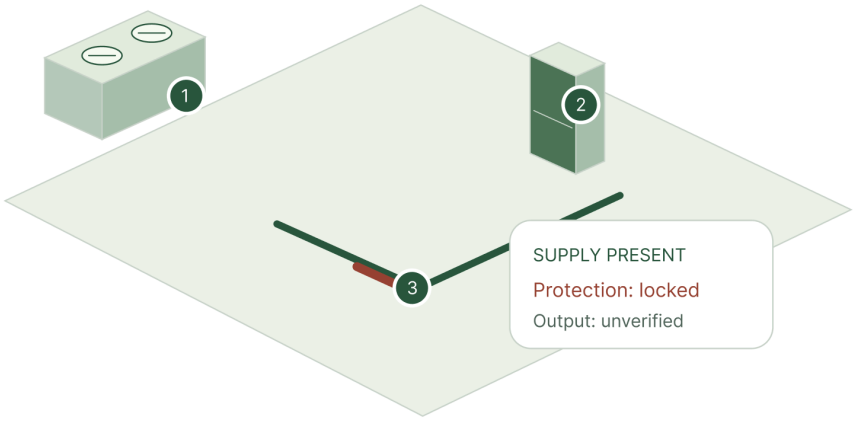
Power remained present. Cooling stayed locked out.

May 29-30, 2026 | West US 2, Washington

Storm-related voltage disturbances put cooling components into protective lockout across facilities in two physical availability zones. Rising temperatures shut down infrastructure. Manual recovery, storage validation and telemetry backlogs prolonged customer impact. Microsoft identified clearer cooling-recovery procedures, visibility into protective states and service-startup improvements.

Electrical supply is only one state.

Control state and useful output need their own evidence.



01 Cooling equipment

Voltage disturbances put components into protective lockout.

02 Control state

Supply remained present; the equipment still required recovery.

03 Verified output

Cooling returned before infrastructure and services fully recovered.

Supply, control state and useful output are separate conditions.

Published mechanism with conceptual architectural artwork; site geometry and equipment counts are illustrative.

91 minutes

From first customer impact to cooling restoration

Protection state changed what recovery required.

Supply availability alone could not restore cooling.

R19: RCA / Video / Essay

CASE 03 / THE RECOVERY STORY

A restored supply is only one recovery condition.

PUBLISHED MILESTONES

- **May 29, 04:24 UTC**
  Customer impact begins.
- **05:55**
  Cooling restored; temperatures stabilized.
- **12:00**
  About 95% of affected underlying VMs recovered.
- **20:18**
  Only two telemetry services remain impacted.
- **May 30, 02:30**
  Final impact mitigated.

Full published impact window: 22h 06m; customer experience varied.

Sequence shown; spacing is not elapsed time.

01 **Make protective states explicit.**

Equipment may be powered, reachable and unable to perform its job. A recovery record should distinguish supply availability, control state, mechanical function and verified output. Those distinctions help responders avoid assuming that a green upstream indication settles the downstream question.

02 **Account for design differences.**

A common portfolio template needs site-specific exceptions. Identify the equipment families, control arrangements and operating sequences that change the response. Standardize how differences are documented while keeping the actual instructions tied to the applicable system.

03 **Track operational recovery separately.**

Facilities, infrastructure and customer support may each use the word restored differently. Agree the evidence for each milestone before an incident. Assign ownership of accumulated work as well as live requests so the final recovery stage remains visible.

TAKE THIS QUESTION TO YOUR TEAM

Can the next shift tell which devices recover automatically and which need an authorized intervention?

R19: RCA | Operating lessons: Entangl analysis.

CASE 03 / WHERE ENTANGL HELPS

Put the correct recovery procedure beside the asset.

Entangl can help prepare and review procedures from the equipment information your team already maintains. This scenario is a useful test of whether a portfolio procedure preserves the differences that matter at a particular facility.

ENTANGL EVALUATION / CONCEPTUAL REVIEW CANVAS

One asset. Four separate checks.

ASSET REFERENCE

Identity
Configuration
Applicable manual

Resolve aliases and source revision.

01 **Supply**

What proves it is present?

02 **Protection**

Which state is reported?

03 **Function**

What output is verified?

04 **Service**

Who accepts recovery?

The review must preserve equipment-specific differences and intervention authority.

Bring: equipment evidence: Relevant manuals, sequences and protection-state descriptions; site evidence: Asset list, aliases and the recovery responsibilities; review evidence: Existing EOP, exception list and prior recovery observations.

Build a recovery reference

Connect equipment identities to the applicable source material. Have the site expert resolve differences between the documented design and observed configuration.

Challenge the procedure

Review the draft for missing state checks, ambiguous handoffs and unsupported assumptions about automatic recovery. Evaluate the clarity of the evidence presented to the engineer.

Train for the exception

Rehearse a device that remains unavailable after upstream conditions normalize. Measure the time to locate the correct reference and identify the responsible specialist.

Implementation focus. A software workflow cannot clear a physical protection state by implication. Device controls, thresholds and interventions remain part of the site engineering and authorization process.

Explore this with Entangl →

Explore the Entangl workflow | Conceptual evaluation; not a product screenshot.

CASE 04 / NEBIUS

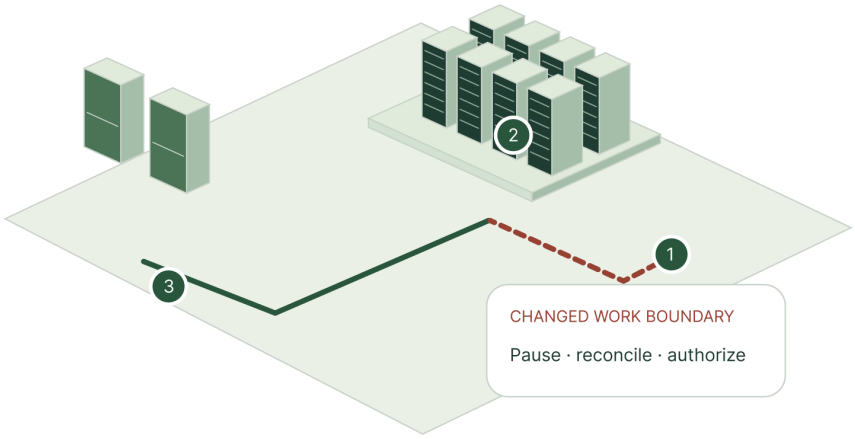
Maintenance crossed its approved boundary.

March 10-11, 2026 | us-central1, United States

Provider-led electrical work exceeded the expected impact. A temporary power configuration, UPS depletion and switching/ATS failures de-energized additional racks. Further interruptions disrupted recovery. Shared regional dependencies prevented some workloads from restarting automatically. Nebius committed to stronger planning, stop-and-review criteria and dependency-ordered recovery drills.

Temporary power changes the boundary.

Review the configuration that exists during the work.



01 Temporary configuration

The electrical arrangement changed during provider-led work.

02 Wider impact

Additional racks lost power beyond the expected scope.

03 Work stopped

Electrical work stopped while service recovery continued.

A changed operating condition needs a new decision about the work.

Published mechanism with conceptual architectural artwork; site geometry and equipment counts are illustrative.

9h 32m

Overall reported impact window across multiple interruptions

The temporary arrangement changed the failure boundary.

The work package needed a clear stop-and-review decision.

R15: RCA

CASE 04 / THE RECOVERY STORY

Treat a scope change as a new decision.

PUBLISHED MILESTONES

- **Mar 10, 15:18 UTC**
  Initial power interruption.
- **17:04**
  Broad external VM connectivity recovers.
- **19:53**
  Further power interruption.
- **23:05**
  Electrical work stops for the night.
- **Mar 11, 00:50**
  Customer impact fully mitigated.

The wider incident contains several distinct service-impact intervals.

Sequence shown; spacing is not elapsed time.

01 **Review the temporary configuration.**

A maintenance plan should describe the arrangement during work, including the period between steps. The normal one-line is a starting point; reviewers also need to understand which protections, supplies and contingencies remain available in each temporary state.

02 **Define what invalidates approval.**

An unexpected interruption, a changed dependency or a missing witness can invalidate the assumptions behind the original plan. Write those conditions as explicit decisions with a responsible role. A warning that nobody is empowered to act on has little operational value.

03 **Separate continuation from recovery.**

An incident response and a maintenance program can compete for the same equipment and people. Establish who decides whether further work can proceed, what evidence is required and how the revised plan is communicated to teams already restoring service.

TAKE THIS QUESTION TO YOUR TEAM

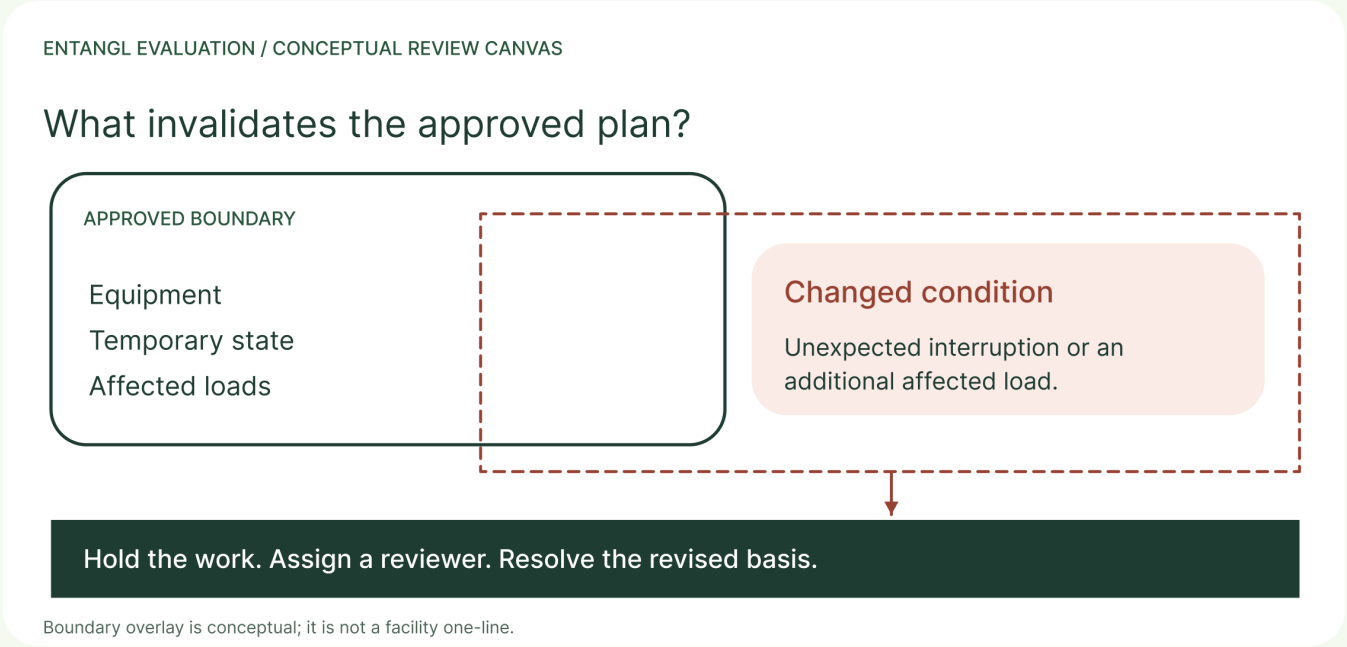
When the work changes, who has the authority and evidence to stop it?

R15: RCA | Operating lessons: Entangl analysis.

CASE 04 / WHERE ENTANGL HELPS

Keep the procedure aligned with the work.

Use Entangl to prepare and review the maintenance package against its site sources, operating requirements and expected format. Focus the evaluation on the conditions that would make the original plan unsafe or incomplete.



Bring: work package: Procedure, contractor scope and maintenance window; site context: Current one-lines, temporary arrangements and asset names; decision rules: Stop conditions, escalation roles and revision requirements.

Inspect the assumptions

Ask reviewers to identify the temporary states on which the procedure relies. Connect each important claim to a source or a required field confirmation.

Make the exception usable

Assess whether the procedure clearly tells the team when to pause, whom to contact and what must be reviewed before continuing. Test a realistic scope-change scenario.

Retain the version that was reviewed

Demonstrate how a changed source or procedure is distinguished from the accepted version in the deployed process. Measure review effort and unresolved ambiguities.

Implementation focus. The operating organization controls work authorization and contractor execution. Entangl supports the preparation and review of those decisions; it cannot guarantee compliance in the field.

Explore this with Entangl →

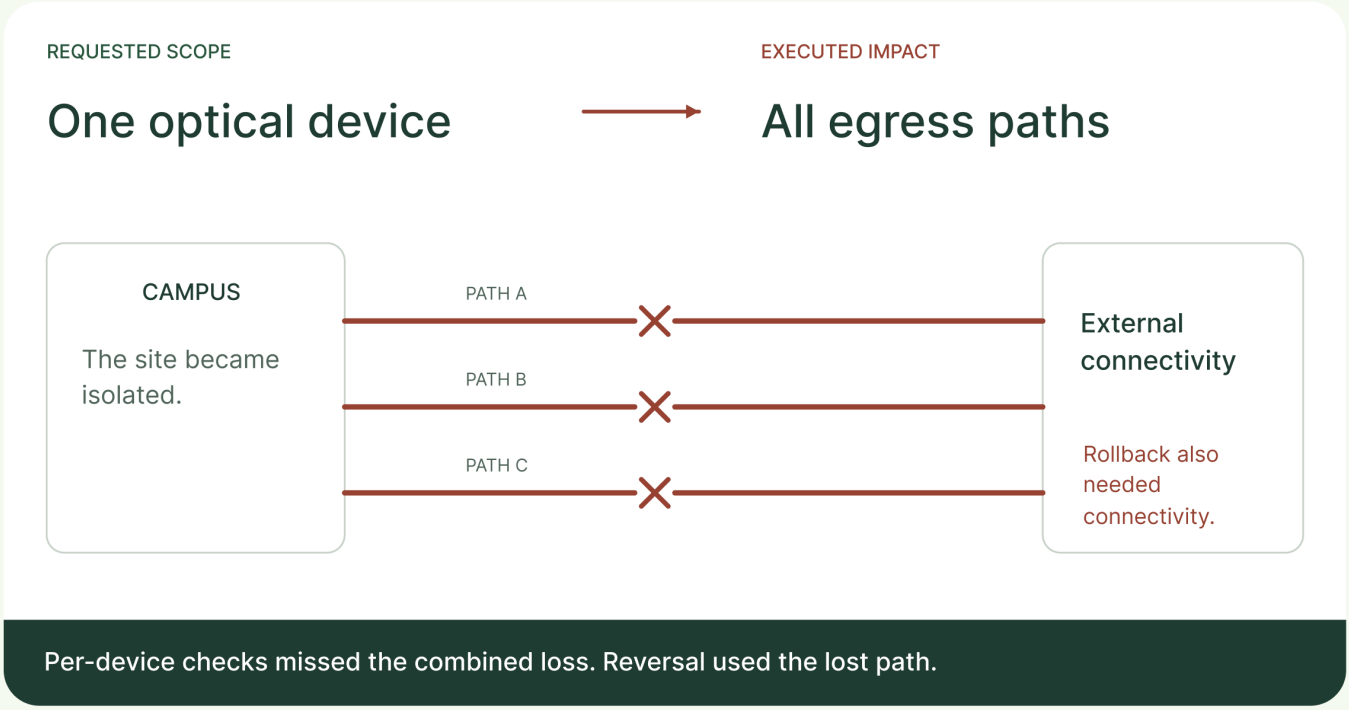
Explore the Entangl workflow | Conceptual evaluation; not a product screenshot.

CASE 05 / MICROSOFT AZURE

Each device passed. The combined change failed.

July 23, 2026 | West US, California

Optical-repair automation expanded a single-device request to all egress devices at a data center. Per-device validation missed the aggregate loss of redundant paths. Physical links appeared healthy, and automated rollback depended on the disrupted connectivity. Microsoft reported scope and path-validation fixes, plus further work on visibility and rollback.



Per-device checks missed the combined loss. Reversal used the lost path.

Mechanism from the public RCA. Paths and equipment positions are schematic; counts are illustrative.

4h 57m

Published service-impact window

Per-device checks missed the combined loss of paths.

Review the entire change and the route needed to reverse it.

R21: RCA / Video / DCD

CASE 05 / THE RECOVERY STORY

Check the combined effect of the change.

PUBLISHED MILESTONES

- **14:44 UTC**
  Routes withdrawn; impact begins.
- **17:19**
  Engineers connect the incident to the repair.
- **17:45**
  Manual rollback starts.
- **18:26**
  Data-center connectivity restored.
- **19:41**
  Affected services fully recovered.

Impact concerned traffic entering or leaving the region.

Sequence shown; spacing is not elapsed time.

01 ### The unit of review matters.

A device-level check answers a device-level question. An operating review must also ask what happens to the service boundary when the full work package is applied. Explicitly define the shared paths, dependencies and capacity that the combined operation can remove.

02 ### Make automated work discoverable.

A change that runs without a person in the loop still needs an identifiable record. Incident responders should be able to connect the affected scope, the initiating request, its source version and the responsible system or owner without reconstructing them from separate logs.

03 ### Protect the route back.

Rollback has prerequisites just as execution does. Review the connectivity, credentials, tooling and people needed to reverse a change. A recovery method that depends on the resource being removed needs an independently available alternative.

TAKE THIS QUESTION TO YOUR TEAM

Does the review test every task separately, or the state left by all of them together?

R21: RCA | Operating lessons: Entangl analysis.

CASE 05 / WHERE ENTANGL HELPS

Put the dependency question into the review.

Entangl can help reviewers inspect a procedure alongside the equipment and source relationships it references. Use the combined-change problem as a review scenario, with the network specialist defining the topology and acceptance criteria.

ENTANGL EVALUATION / CONCEPTUAL REVIEW CANVAS

The batch changes the answer.

SURVIVING PATH	CHANGE 1	CHANGE 2	COMBINED
Path A	AFFECTED	AVAILABLE	AFFECTED
Path B	AVAILABLE	AFFECTED	AFFECTED

Review the combined impact and the connectivity needed to reverse it.

Illustrative two-path review example; these are not the incident device counts.

Bring: change scope: All included targets and the intended service boundary; dependency evidence: Validated topology and the paths required during work; recovery evidence: Rollback references, access requirements and escalation owner.

Make the scope inspectable

Compare the assets named in the work package with the documented system boundary. Require the specialist to resolve aliases, omissions and any broader target set.

Review prerequisites and contingencies

Ask whether the procedure provides evidence for surviving connectivity and for the return path. Assess useful findings, missed concerns and unnecessary flags.

Rehearse loss of normal access

Have the team walk through a case in which the usual management path is unavailable. Record what additional engineering or tooling is needed before execution.

Implementation focus. This is a procedure and evidence review application. Network simulation, route validation and enforcement inside a provider automation system require specialist tools and integration.

Explore this with Entangl →

Explore the Entangl workflow | Conceptual evaluation; not a product screenshot.

CASE 06 / GOOGLE CLOUD

The missing instruction was the sequence.

September 1, 2026 | Iowa: portions of us-central1-b and -f

An upgrade work order omitted the intended one-router-at-a-time sequence. A technician replaced incompatible optical components across redundant paths; expected validation and stop safeguards failed. The final RCA reports 4h 11m of impact to affected clusters, not the entire region. Google committed to automatic sequencing, work-stop alerts, training and faster traffic diversion.

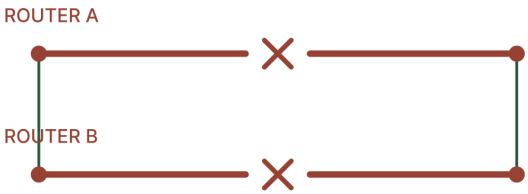
INTENDED SEQUENCE



One path remains available.

One router at a time.

ISSUED WORK ORDER



Both redundant paths disconnected.

That ordering instruction was omitted.

The field version lost the sequence that preserved redundancy.

Mechanism from the public RCA. Paths and equipment positions are schematic; counts are illustrative.

13 minutes

Time in which redundant fiber paths were unplugged

The issued work order lost the intended sequence.

The field instruction must carry the operating logic.

R31: RCA / The Register

CASE 06 / THE RECOVERY STORY

The handoff must carry the operating logic.

PUBLISHED MILESTONES

- **07:41 Pacific**
  Final RCA impact window begins.
- **08:00**
  Regional traffic diversions in place.
- **11:52**
  Final RCA impact window ends.
- **September 10**
  Final RCA published.

The status header starts three minutes later than the final RCA.

Sequence shown; spacing is not elapsed time.

01 **A list of tasks is not an executable sequence.**

Order, prerequisites and release criteria are part of the instruction. Review the actual package the technician receives, including attachments and work-system exports. A complete engineering plan can still become an incomplete field instruction during handoff.

02 **Make validation a step with evidence.**

A request to check that everything is healthy needs a defined observation and an accountable role. Specify what the team must establish before the next stage can begin, using the site-approved method and a record the supervisor can inspect.

03 **Test the procedure with a fresh reader.**

A person unfamiliar with its authoring history is more likely to expose missing context. During a dry run, ask the reader to explain the next permitted action, the condition that allows it and the point at which they would stop and escalate.

TAKE THIS QUESTION TO YOUR TEAM

Can someone execute the issued work order correctly without relying on unwritten knowledge?

R31: RCA | Operating lessons: Entangl analysis.

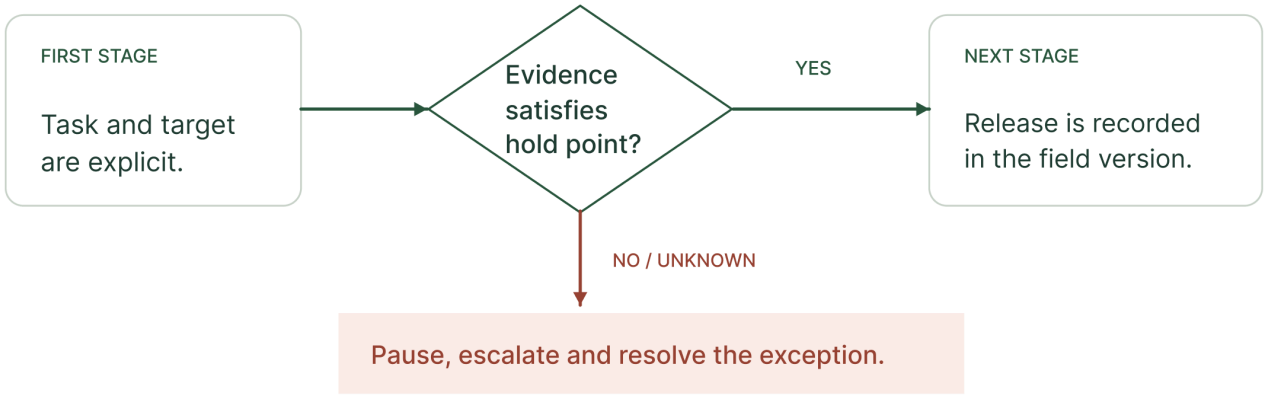
CASE 06 / WHERE ENTANGL HELPS

Review what the technician will actually receive.

Entangl supports procedure preparation and review using the site information and requirements your team supplies. Combine that with a training dry run to test whether the issued work package preserves the operating sequence.

ENTANGL EVALUATION / CONCEPTUAL REVIEW CANVAS

A release condition is part of the instruction.



Test the actual issued version with a fresh reader. A complete authoring draft is insufficient.

Bring: field version: The document and attachments used at the equipment; technical context: Equipment identities, dependencies and approved constraints; acceptance examples: A known-good procedure and a deliberately incomplete sequence.

Assess the instructions

Review dependencies, prerequisites and the ordering of steps. Ask the engineer to evaluate each finding against the task and its cited evidence.

Walk the exception path

Use a rehearsal to expose ambiguous equipment names, missing observations or a failed prerequisite. The practice should show when the team pauses and how the issue reaches the reviewer.

Measure understanding as well as speed

Track unresolved questions, meaningful review findings and the time to reach an accepted field-ready document. Confirm that the exported version preserves the agreed sequence.

Implementation focus. An approved procedure still depends on competent execution and local authorization. Entangl does not establish optical compatibility or automatically enforce a provider network workflow.

Explore this with Entangl →

Explore the Entangl workflow | Conceptual evaluation; not a product screenshot.

CASE 07 / GOOGLE CLOUD

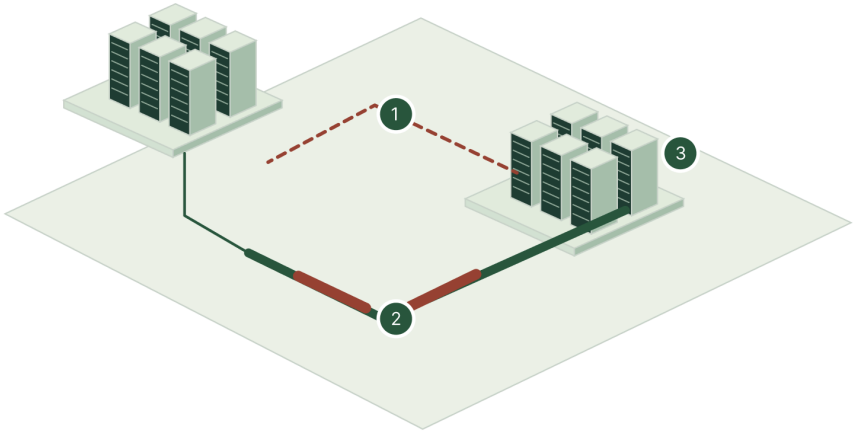
Alternate paths existed. Traffic still congested.

August 20, 2026 | us-west1, Oregon

Planned fiber work reduced capacity between data centers. Automatic rerouting did not redistribute traffic adequately, causing congestion and cascading service failures. Some backlog processing and state reconciliation continued after core recovery. Google identified stricter maintenance checks, circuit redundancy requirements and improved traffic failover as prevention priorities.

The remaining path must carry the load.

A route can remain connected and still be inadequate.



01 Capacity removed

Planned fiber work reduced inter-data-center capacity.

02 Traffic concentrates

Rerouting did not redistribute traffic adequately.

03 Service aftermath

Congestion cascaded into service failures and residual recovery.

An alternate path must support the traffic that needs to move.

Published mechanism with conceptual architectural artwork; site geometry and equipment counts are illustrative.

2h 22m

Core impact window in the final RCA

The remaining paths could not absorb traffic as intended.

A documented alternative still needs validated capacity.

R30: RCA

CASE 07 / THE RECOVERY STORY

A second path needs usable headroom.

PUBLISHED MILESTONES

- **08:00 Pacific**
  Core impact begins in final RCA.
- **10:22**
  Core connectivity and request serving restored.
- **After 10:22**
  Some backlogs and reconciliation continue.
- **August 27**
  Final RCA published.

The status header uses a different interval from the final RCA.

Sequence shown; spacing is not elapsed time.

01 **Connectivity and capacity answer different questions.**

A documented alternative establishes a route to investigate, not proof that the route can sustain the intended service. Before work, identify the load assumptions, the engineering evidence for available capacity and the operating conditions under which those assumptions must be reconsidered.

02 **Review the transition, not only the endpoint.**

Traffic may concentrate unevenly while a network changes state. The work package should reference the approved diversion plan, validation points and conditions for pausing. The network team must define and verify the technical behavior; the procedure must carry those decisions accurately.

03 **Include the dependent services.**

A short network disruption can leave downstream state or work queues that need attention. Ask service owners what remains to be checked after connectivity returns and who owns each residual recovery task. That prevents infrastructure completion from silently becoming a customer-restoration claim.

TAKE THIS QUESTION TO YOUR TEAM

Have you tested the capacity available during maintenance, including the traffic that must move?

R30: RCA | Operating lessons: Entangl analysis.

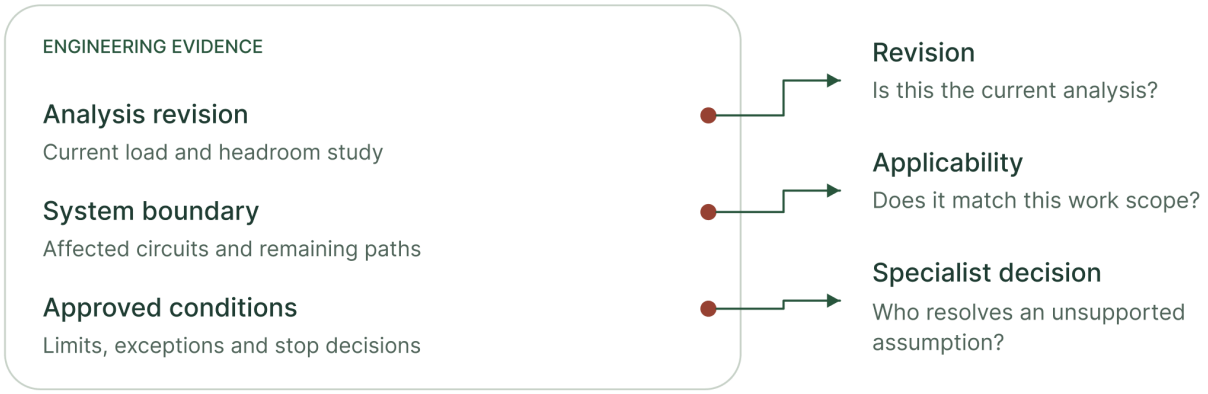
CASE 07 / WHERE ENTANGL HELPS

Put capacity assumptions in the review.

Use Entangl to connect the maintenance procedure to the source information and specialist approvals that justify it. The objective is to expose an unsupported assumption before a change proceeds, then retain the engineer's decision with the reviewed version.

ENTANGL EVALUATION / CONCEPTUAL REVIEW CANVAS

Make the capacity assumption inspectable.



The procedure cites the analysis; engineering tools and specialists establish capacity.

Bring: work scope: Affected circuits, timing and the diversion procedure; engineering evidence: Approved capacity analysis and operating limits; service evidence: Dependent services, acceptance tests and recovery owners.

Review the source set

Assess whether the procedure references the right analysis, revision and system boundary. A missing capacity study should remain an explicit question for the responsible engineer.

Check the decision points

Inspect prerequisites, validation steps, escalation and rollback references. Rehearse a result that falls outside the approved assumptions and identify who decides what happens next.

Evaluate the review outcome

Measure whether meaningful gaps reach the correct specialist, how quickly they are resolved and whether the accepted work package retains the supporting evidence.

Implementation focus. Entangl is not being represented as a network capacity simulator or traffic-engineering controller. Those calculations and validations remain with the engineering systems and specialists.

Explore this with Entangl →

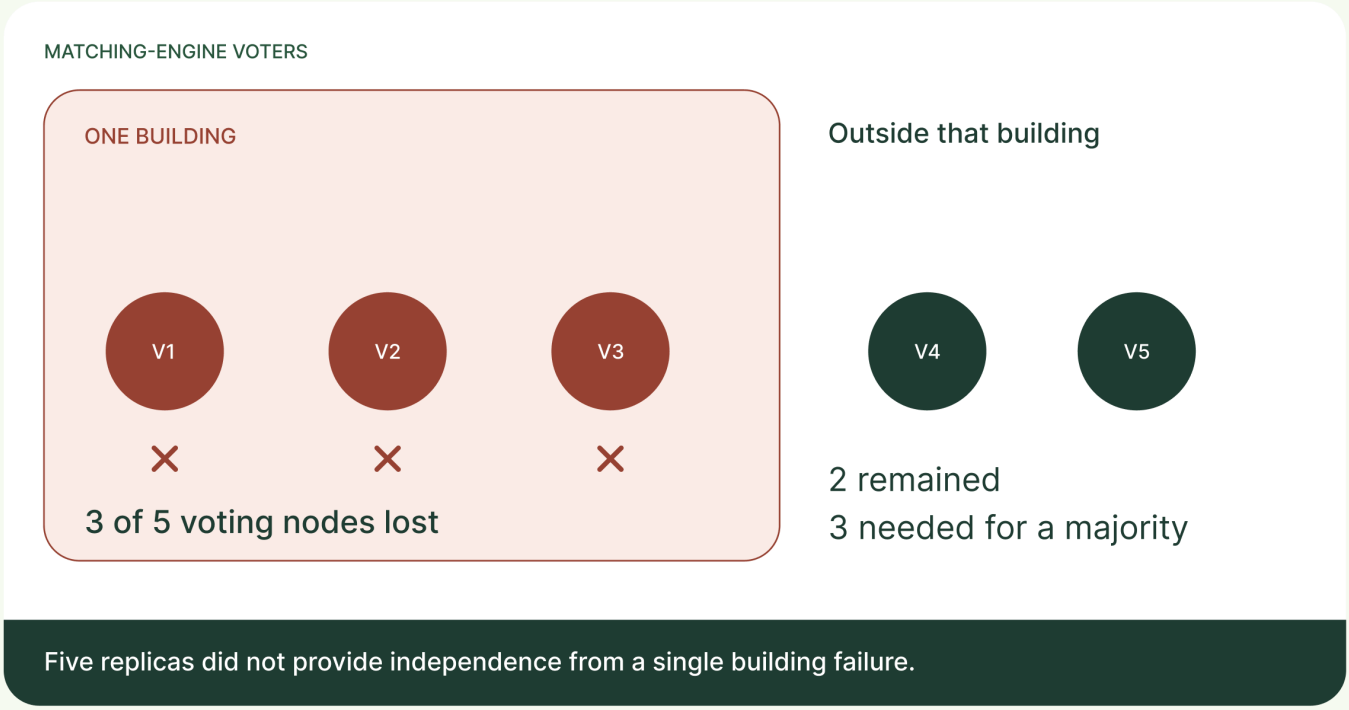
Explore the Entangl workflow | Conceptual evaluation; not a product screenshot.

CASE 08 / AWS / COINBASE

One building failed. Trading stopped.

May 7-8, 2026 | Northern Virginia: AWS us-east-1

Coinbase attributes the event to simultaneous AWS data-hall chiller failures. Its matching engine lost three of five voting nodes in one building, with no automatic cross-zone failover. A startup-code change was needed. A managed Kafka control-plane defect also blocked recovery. Coinbase committed to warm standby, failover exercises and Kafka recovery tooling.



Voting-node counts follow the customer RCA. The building outline and node positions are schematic.

About 8 hours

Severe Coinbase disruption, followed by further recovery

Replication did not provide building-level independence.

Facility recovery and application failover have distinct owners.

R18: RCA / DCD / Essay

CASE 08 / THE RECOVERY STORY

Define responsibility before the incident.

PUBLISHED MILESTONES

- **May 7, 19:20 Eastern**
  Reported chiller failures.
- **May 8, 02:25**
  Matching engine opens in cancel-only mode.
- **03:49**
  Full order-book trading resumes.
- **09:53**
  Retail experience fully recovered.
- **14:00**
  Remaining streaming backlog clears.

Detailed customer RCA; equivalent public AWS facility RCA not located.

Sequence shown; spacing is not elapsed time.

01 **Count independent failure domains.**

Replica counts do not describe physical independence. Ask each service owner to explain the placement assumptions behind availability and the tradeoffs behind them. Where latency or state requirements constrain placement, make the recovery mechanism and its owner explicit.

02 **Document what the provider must do.**

A managed service can still have a failure condition that requires provider engineering. Build a joint escalation path with the information needed for a useful handoff. Distinguish actions available to your team from actions that need the provider's access or tooling.

03 **Prove the business operation.**

A recovered component is one milestone. The acceptance test should include the transaction the customer needs to complete and the dependent services required to support it. Rehearse how service owners communicate readiness without assuming that another team's green status closes the whole incident.

TAKE THIS QUESTION TO YOUR TEAM

Can your application recover from losing the building in which its replicas sit?

R18: RCA | Operating lessons: Entangl analysis.

CASE 08 / WHERE ENTANGL HELPS

Make the joint response clear before the call starts.

Entangl's most relevant role here is the operating information and procedure layer around a facility incident. Use it to prepare a shared response package that makes the affected equipment, escalation responsibilities and handoff evidence easier to review.

ENTANGL EVALUATION / CONCEPTUAL REVIEW CANVAS

Every handoff has a sender and a receiver.

FACILITY OPERATOR

Physical service

Affected assets and restoration evidence

CLOUD / PLATFORM

Shared infrastructure

Access, dependencies and provider actions

APPLICATION OWNER

Customer operation

Transaction acceptance and remaining backlog

Record the receiving owner and the evidence required to accept each milestone.

A facility procedure supports coordination; it does not replace application recovery engineering.

Bring: facility interface: Affected-site context and operator escalation contacts; response package: Facility EOPs and the service teams they must inform; recovery agreement: Milestones, evidence and owners for each handoff.

Connect the operating references

Prepare the facility-side response with the current documents and named roles. Identify what the downstream team needs to know to assess its own exposure.

Review the boundary

Check that the procedure distinguishes facility actions from provider and application actions. Retain unresolved dependencies for the team that can actually resolve them.

Rehearse the coordination

Run a tabletop with facility and service owners. Measure time to a complete handoff, missing escalation information and clarity of the final service acceptance decision.

Implementation focus. Entangl does not replace application failover architecture, consensus recovery or a managed-service software fix. Those engineering changes are essential and outside this proposed facility workflow.

Explore this with Entangl →

Explore the Entangl workflow | Conceptual evaluation; not a product screenshot.

CASE 09 / BLACKSMITH

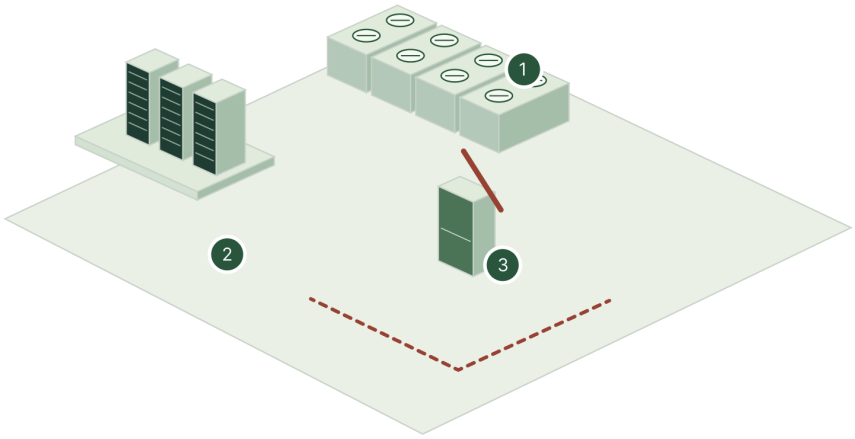
Compute could recover. Storage still blocked work.

August 13-14, 2026 | Phoenix, Arizona

Blacksmith reports utility interruptions disabled all four chillers. Storage lost quorum, unavailable caches stalled jobs, and heat disabled remote management. Blacksmith committed to spreading capacity across facilities, allowing work to proceed without unavailable caches and making mitigation settings adjustable without a deployment.

Restore state and management access.

Compute cannot recover in isolation from its prerequisites.



01 **Cooling lost**

Utility interruptions disabled all four chillers.

02 **State unavailable**

Storage lost quorum; unavailable caches stalled jobs.

03 **Access lost**

Heat also disabled remote management needed for recovery.

Returning compute still depended on state and management access.

Published mechanism with conceptual architectural artwork; site geometry and equipment counts are illustrative.

About 2,400

Hosts unavailable at the reported peak

Cooling loss removed several recovery prerequisites.

Management access and storage can hold up returning compute.

R25: RCA / Essay

CASE 09 / THE RECOVERY STORY

Make degraded operation an explicit design choice.

PUBLISHED MILESTONES

- **Aug 13, about 06:00 UTC**
  Hall temperatures begin rising.
- **09:37**
  NVMe temperature alert.
- **21:45**
  Job pickup near normal.
- **Aug 14, 00:00**
  Remote power control returns.
- **18:09**
  Caching and sticky disks restored.

Some recent cache data was unrecoverable. Facility operator unnamed; possible overlap with Namecheap's same-day Phoenix event.

Sequence shown; spacing is not elapsed time.

01 **Identify the minimum useful service.**

Ask what customers can still accomplish when a nonessential dependency is unavailable. The answer is a product and engineering decision: define what can be disabled, which behavior is acceptable and who can authorize the change. Capture it before the urgent tradeoff arrives.

02 **Include management infrastructure.**

Remote access can be a hidden recovery dependency. Trace its power, network and authentication requirements with the same care as the production service. Then rehearse the circumstances in which local people, alternate access or another facility are required.

03 **Plan the receiving side of failover.**

Moving work changes the load elsewhere. Ask receiving teams what they can accept, what customers will experience and what must happen to stateful dependencies. A useful failover plan includes the new queue and the operator who owns it, not simply a destination name.

TAKE THIS QUESTION TO YOUR TEAM

Which dependency could stop useful work even after your servers begin returning?

R25: RCA | Operating lessons: Entangl analysis.

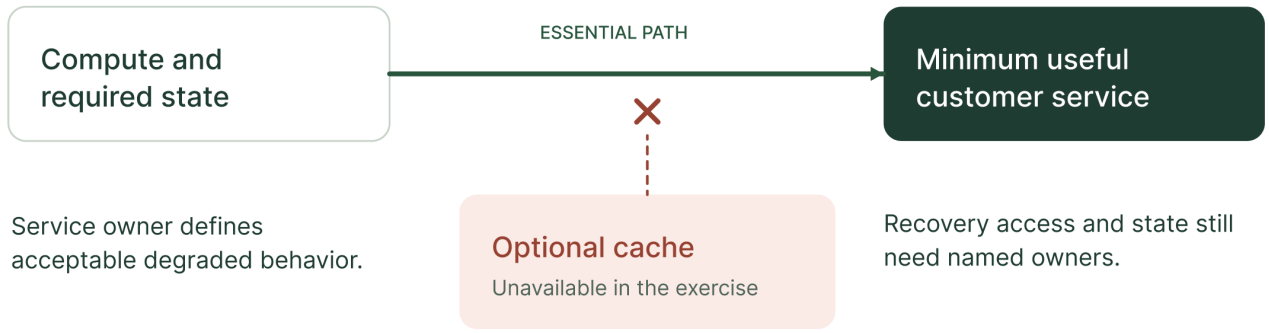
CASE 09 / WHERE ENTANGL HELPS

Review the EOP around the work that must continue.

Entangl can help the facility team prepare, review and rehearse its part of a degraded-service response. Bring the recovery access requirements and service-owner handoffs into the same operating package as the physical incident procedure.

ENTANGL EVALUATION / CONCEPTUAL REVIEW CANVAS

Separate essential work from optional dependencies.



Review the decision and handoff. Implementing a bypass requires application engineering.

Bring: facility documents: Cooling references, management dependencies and contact paths; service decisions: The customer functions to preserve and their responsible owners; recovery evidence: Required state, access and acceptance criteria for each handoff.

Expose the dependencies

Map the facility assets and source references needed to investigate the incident. Ask the team to identify the management systems that would also become unavailable.

Review the decision record

Prepare the operating procedure with the approved escalation routes and service-owner decisions. Make clear which actions are facility operations and which require software teams.

Rehearse the constrained response

Walk through loss of remote access and a reduced-capacity destination. Measure how quickly the team finds the correct reference, owner and workable next step.

Implementation focus. Graceful degradation, cache circuit breakers and workload routing require application engineering. Entangl can support the coordinated operating workflow; it is not a substitute for those changes.

Explore this with Entangl →

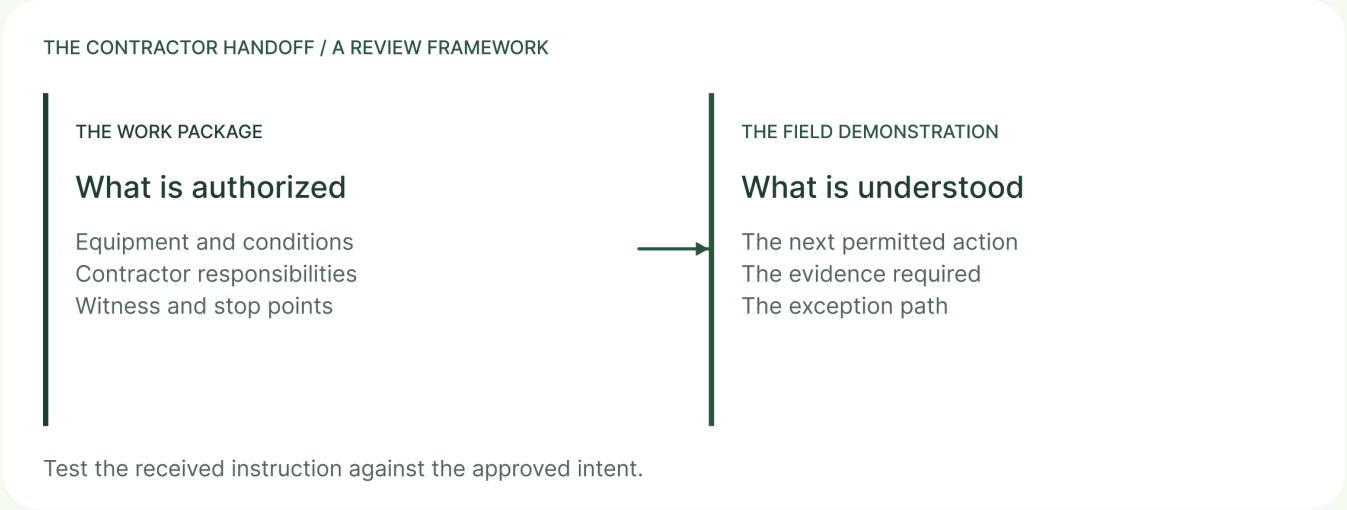
Explore the Entangl workflow | Conceptual evaluation; not a product screenshot.

ADDITIONAL CASE / LIMITED PUBLIC RCA

CME and CyrusOne: the procedure in the field.

Aurora, Illinois | November 27-28, 2025

What is public. Cooling problems at CyrusOne CHI1 interrupted CME trading for roughly ten hours. CyrusOne later attributed the failure to cooling procedures not being followed. A complete public engineering RCA was not located.



Make the handoff executable

Identify the equipment, operating conditions, contractor responsibilities, witness points and completion evidence.

Test understanding

Ask the person receiving the work to explain the prerequisites, exception path and basis for declaring completion.

Where Entangl helps. Use Entangl to prepare or review a seasonal procedure against the applicable equipment documentation and your site requirements. Then use a dry run to test the clarity of the handoff between the author, contractor and supervisor.

What to measure. Compare unresolved questions, missing source references, meaningful findings and total effort to reach an accepted procedure. Keep the technical commissioning or equipment test within the responsible engineering process.

R04: Company/status / Bloomberg / Coverage | Entangl review

ADDITIONAL CASES / LIMITED DISCLOSURE

Oracle and TikTok: responding with partial facts.

United States | January and March 2026

25 JAN

January 25. Oracle attributed a U.S. data-center disruption to weather-related power loss. TikTok problems continued as technical recovery progressed. The exact facility and electrical failure sequence were not disclosed in the sources found.

03-04 MAR

March 3-4. A separate Ashburn, Virginia disruption involved connectivity, timeouts and latency. The initiating mechanism was not publicly established. It should not be described as another power failure merely because January involved one.

Observed

Customer impact and its scope.

Confirmed

Provider facts and their sources.

Requested

The next evidence and owner.

Where Entangl helps. Apply Entangl to the facility-side response package: relevant site information, an approved EOP, escalation roles and the records needed by the receiving team. Review the procedure for an unknown-cause incident, rather than assuming a diagnosis.

What to measure. In a tabletop, assess how quickly the team finds the correct references and owner, how clearly uncertainty is retained, and whether the handoff gives the provider a useful question to answer.

KEEP UNCERTAINTY VISIBLE

An incomplete diagnosis should produce a clear next question.

R10: Reuters/CNA / Coverage | R14: Company/status / The Verge / Coverage | Entangl procedures

ADDITIONAL CASE / COMPANY EXPLANATION

Namecheap: prove each service is back.

RadiusDC, Phoenix, Arizona | August 13-14, 2026

What is public. Namecheap's CEO described storm-related cooling failure, protective shutdowns and staged recovery after temporary cooling was installed. Hosting, email and other functions were affected while DNS resolution continued. Namecheap identified RadiusDC as the operator. A detailed facility RCA was not located.

DNS resolution	Continued
Hosting and email	Affected
Other functions	Staged recovery

Service conditions described in the company account, not a scaled outage timeline.

This report may concern the same physical event as Blacksmith's Phoenix account. The relationship is unconfirmed, so the two accounts are retained as reports rather than asserted to be separate facility outages.

Where Entangl helps. Use Entangl to prepare and review the operating procedure around a staged restoration. Include the evidence each service owner needs, the handoff to customer communications and the conditions that keep an unresolved item open.

What to measure. Rehearse a mixed state in which one customer function is healthy and another is not. Assess whether each claim of recovery has an owner, evidence and a timestamp that the receiving team can interpret.

THE OPERATING LESSON

Define restored for each service.

R24: Company/status / TechRadar | Entangl procedures

ADDITIONAL REPORTS / GPU INFRASTRUCTURE

CoreWeave: look beyond node counts.

Four public reports | August-September 2026

01 / Power	02 / Host	03 / Fabric	04 / Workload
Facility condition	Machine availability	GPU communication	Customer operation

Aug 3 / US-EAST-04A	Power issue took a group of nodes offline. Electrical mechanism undisclosed.
Aug 4 / US-EAST-04A	Networking hardware issue affected nodes. Relationship to Aug 3 unconfirmed.
Aug 19 / DH204	Node failures reduced capacity in 19 GB300 racks. Actual recovery duration not established.
Sep 3 / US-CENTRAL-08A	Backend GPU fabric degradation. Degradation does not establish total outage.

The operating question. Can the team distinguish physical power, host availability, placement capacity and interconnect performance? A count of available machines can hide the condition that prevents a particular workload from running.

Where Entangl helps. Use the site model and operating documents to give the facility team consistent equipment names and dependencies. Review an incident handoff that separates what facilities can verify from what cluster and application teams must investigate.

A useful evaluation. Bring a historical incident and the facility records available at the time. Measure the quality of the equipment mapping, missing context and the clarity of the next responsible team. Test how the workflow handles unknown cause and partial degradation.

Individual operator and archive links: R22, R23, R29 and R32 on page 50. | Entangl telemetry

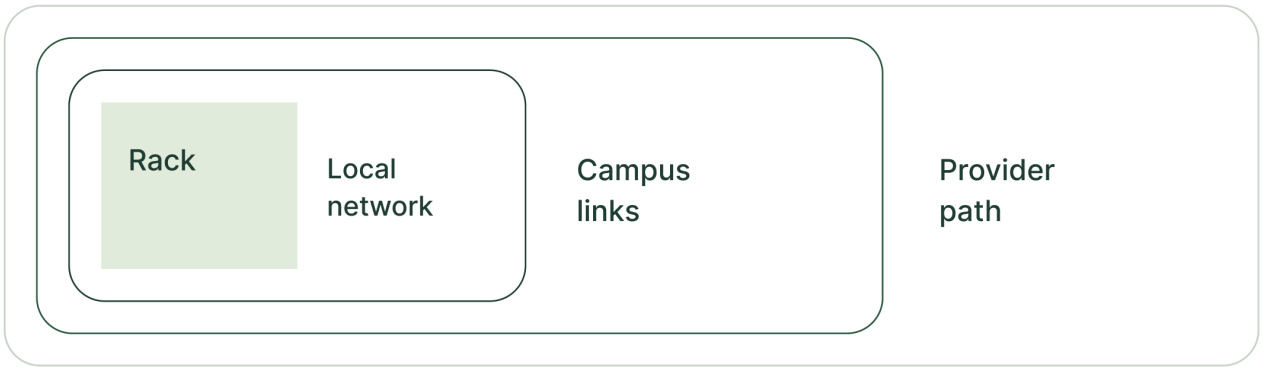
ADDITIONAL REPORTS / LOCAL NETWORKS

Equinix Metal: make the handoff specific.

Ten incident reports | September 2025-August 2026

What the record contains. The register includes New York switch and backbone incidents, DC-metro network instability, Silicon Valley connectivity incidents, a Chicago device restart, Atlanta network isolation and a brief two-metro DDoS event. Most provide status updates rather than a full public RCA.

LOCATE THE FAILURE DOMAIN / DO NOT INFER THE CAUSE FROM ONE SYMPTOM



Confirm the affected layer, its source evidence and the operator responsible for it.

An effective handoff does not require a premature diagnosis. It requires the affected equipment or service, the observed symptom, timestamps, relevant changes and a clear request for investigation. Evidence that narrows the boundary is useful even while the initiating cause remains unknown.

Where Entangl helps. Relate the equipment named in a procedure or investigation to the site information your team can validate. Prepare a network-incident EOP with the right contact, source references and evidence requirements for each handoff.

A useful evaluation. Rehearse two similar customer symptoms with different affected boundaries. Measure whether the response directs a clear, evidence-backed request to the correct team. Specialist network diagnosis and attack mitigation remain with the responsible systems and engineers.

Individual operator and archive links for ten Equinix reports: pages 49-50. | Entangl site-model

VERIFICATION LEADS

Keep the unknowns visible in the record.

Useful research includes knowing what a source does not establish.

Alaska Airlines / Oct 23

Company-reported primary data center failure.

OPEN: The U.S. location of the facility is unconfirmed.

OVH / Dec 15 and 18

Archived Vint Hill rack-power incident reports.

OPEN: Original operator notices were not independently recovered.

OVH / Apr 2-3

Hillsboro rack management-access disruption.

OPEN: Production-traffic outage is not established.

OVH / Aug 15-16

Archived Vint Hill rack-connectivity report.

OPEN: Dates and timing require clarification.

These items are retained in the source register so readers can inspect the evidence. They are not used to claim additional confirmed, site-wide U.S. failures or to support a precise count of distinct facilities.

The operating lesson. The same discipline belongs in a site investigation. A drawing, an alarm and a colleague's account establish different things. State the observation, preserve its source and give the unresolved question to someone who can answer it.

Where Entangl helps. Site-context and procedure review can make missing references and unresolved equipment relationships part of the work the engineer assesses. In an evaluation, deliberately supply an incomplete source set and check that the gap remains visible.

Evidence and source links: R03, R06, R07, R16 and R26 on pages 49-50. | Entangl site-model

OPERATOR WORKSHEET / READINESS REVIEW

Ten questions for your next operating review.

Record the evidence, owner and next action. A confident answer without evidence is still open.

- [ ] **1. What survives the full maintenance scope?**

  Temporary configuration and validated remaining paths.

  OWNER / NEXT ACTION

- [ ] **2. What would invalidate the original approval?**

  Stop conditions and the authorized escalation role.

  OWNER / NEXT ACTION

- [ ] **3. Does the field package preserve the sequence?**

  Issued procedure, attachments and release points.

  OWNER / NEXT ACTION

- [ ] **4. Which states require a different response?**

  Equipment-specific sources and observed configuration.

  OWNER / NEXT ACTION

- [ ] **5. What remains visible if local management fails?**

  Monitoring and communication dependencies.

  OWNER / NEXT ACTION

- [ ] **6. Who owns the first credible warning?**

  Acknowledgement, investigation and handoff responsibilities.

  OWNER / NEXT ACTION

- [ ] **7. How do teams regain management access?**

  Access prerequisites and the approved alternative.

  OWNER / NEXT ACTION

- [ ] **8. What must recover before each dependent service?**

  A reviewed restoration sequence and acceptance criteria.

  OWNER / NEXT ACTION

- [ ] **9. What useful work can continue in degraded mode?**

  Approved service-owner decisions and limitations.

  OWNER / NEXT ACTION

- [ ] **10. What proves the customer has recovered?**

  A transaction or workload test, plus backlog ownership.

  OWNER / NEXT ACTION

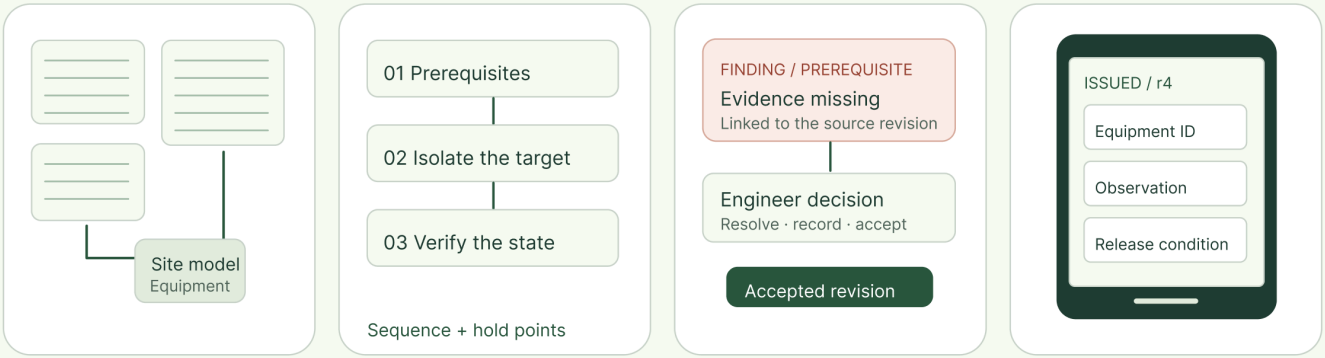
Choose the weakest answer. Turn it into one procedure, evidence or rehearsal task your team can finish and review.

Put the readiness review to work with Entangl

OPERATOR WORKSHOP / MAINTENANCE

Review one MOP from source to field.

A focused procedure review produces something the team can improve and use.



01 / Sources
Assemble the evidence

02 / Preparation
Build the sequence

03 / Review
Resolve the finding

04 / Field version
Verify the handoff

01 ## Choose one task

Bring a familiar maintenance procedure, its field version, template, drawings, asset list and applicable equipment instructions.

02 ## Set the review questions

Identify the temporary states, prerequisites, release conditions, stop decisions and contingency references the work depends on.

03 ## Prepare and review

Use Entangl with the relevant site context. Inspect each finding, its source and the engineer's decision to revise, clarify or accept.

04 ## Test the handoff

Have a fresh reader explain the next permitted action. Introduce a changed drawing or unexpected observation and inspect the exception path.

05 ## Close the loop

Retain the accepted version and decisions. Give unresolved items owners; document completion is not work authorization.

06 ## Measure the result

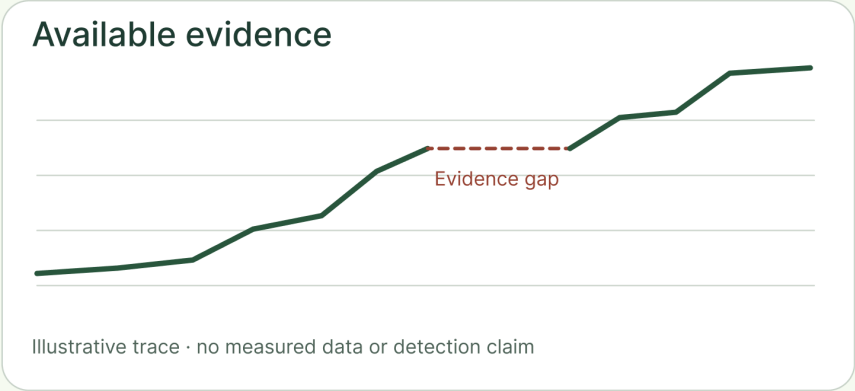
Compare total effort to accepted procedure, meaningful findings, missed issues, false positives and unresolved questions.

Entangl review | Conceptual editorial illustration.

OPERATOR WORKSHOP / INCIDENT RESPONSE

Replay a cooling incident with the real evidence.

Start with a historical event your team understands. Compare the investigation, not a polished demonstration.



Available evidence. Equipment context. A supported next investigation.

01 **Assemble the record**

Bring timestamps, alarms, trends, equipment mapping and the actual response. Mark missing evidence instead of supplying hindsight.

02 **Choose the decision**

Select a useful endpoint: an assigned investigation, the correct EOP, an operator escalation or a specialist handoff.

03 **Replay with Entangl**

Inspect the relationship between available signals, equipment and sources. Ask which explanation is supported and what remains uncertain.

04 **Test difficult evidence**

Include a stale reading, missing point or ambiguous asset name. Check whether the gap changes the proposed next verification.

05 **Agree the deployment**

Confirm the connection, point mapping, timestamps and operator handoff. Define any control functions through their own engineering process.

06 **Measure the result**

Compare investigation time, useful context, missed evidence, false correlations and handling of stale or absent data.

Entangl telemetry | Conceptual editorial illustration.

OPERATOR WORKSHOP / RECOVERY

Rehearse the return of the service.

A recovery exercise should expose the dependencies between teams before those teams meet under pressure.

RECOVERY REHEARSAL / VALIDATE THE HANDOFF

01	02	03	04
Physical conditions	**Foundations**	**Customer operation**	**Accumulated work**
Supply and cooling	Access, storage and network	Transaction or job	Queues and reconciliation
Facilities	Infrastructure	Service owner	Backlog owner

Remove normal access. Introduce a failed restart. Ask what permits the next stage.

01 **Start at the service**

Define the customer transaction or workload. Trace its infrastructure, access, state and decisions to an accountable team.

02 **Prepare the procedure**

Use Entangl to organize and review facility recovery references and service-owner handoffs. Assign unknowns for resolution.

03 **Walk a constrained case**

Remove normal management access. Introduce a component that does not restart automatically. Ask what permits the next stage.

04 **Observe coordination**

Record time spent finding information, waiting for an owner, repeating an action and deciding whether a prerequisite is satisfied.

05 **Validate each endpoint**

Confirm physical conditions, foundational services, customer operations and backlog clearance with the receiving owner.

06 **Measure the result**

Track unresolved dependencies, ambiguous decisions, missing escalation information and the evidence behind each recovery milestone.

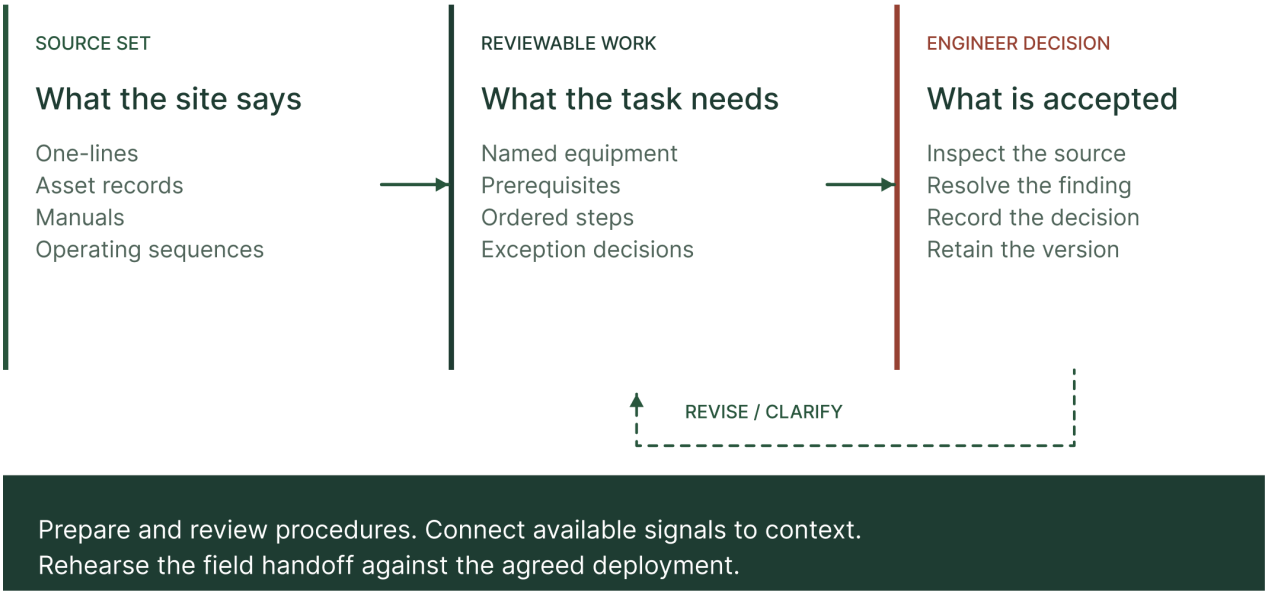
Entangl fieldwork

ENTANGL / PLATFORM FIT

Your site information. Put to work.

Entangl is an AI platform for data center engineering and operations. Start where your team has a recurring task and a clear way to judge the result.

A SITE-CONTEXT WORKFLOW / CONCEPTUAL EVALUATION, NOT A PRODUCT SCREEN



Start with a recurring task

Procedure preparation, review and site context are practical entry points.

Show the actual workflow

Demonstrate telemetry and field handoffs against the proposed deployment.

Let engineers judge

Agree source requirements, responsibilities and acceptance criteria before expanding.

Include information access, processing location, integrations and deployment requirements in the evaluation. The agreed scope should describe the work you will use and the people responsible for it.

Product scope: procedures / review / site-model / telemetry / fieldwork

THE BUSINESS CASE

Measure the work you are improving.

A credible operating case starts with a baseline. It does not require guessing that a named outage would have been avoided.

Time

Total effort to an accepted result.

Quality

Meaningful findings, misses and false positives.

Readiness

Open dependencies and ambiguous decisions.

Build the value estimate from your own work.

Net hours saved per accepted task

x annual task volume = estimated capacity released

Include input preparation, review and correction time.

Use a matched task. Compare work of similar complexity with the same acceptance criteria. Include the effort to prepare inputs and resolve review findings, so a faster first draft does not hide additional downstream work.

Keep time and risk evidence separate. Measured engineering hours can support a productivity case. Review findings and exercise outcomes can show stronger operating preparation. Neither proves a particular reduction in outage probability or insurance cost.

Calculate with your own inputs. Net hours saved per accepted task multiplied by annual task volume gives a workload estimate. If you value that time financially, use your organization's agreed cost assumptions and keep capacity released distinct from cash savings.

Expand after the result is clear. Choose the next site or procedure family because the first evaluation established a useful result, a repeatable source set and an owner for adoption.

Define an evaluation with Entangl

A USEFUL FIRST CONVERSATION

Bring a procedure. Ask the hard questions.

Book an Entangl walkthrough built around work your team already understands.

YOUR FIRST EVALUATION / THREE CHOICES

01

A familiar task

A slow procedure review, an EOP with unwritten knowledge, or an incident with missing context.

02

A useful result

A supported finding, a clearer prerequisite, or a more complete investigation handoff.

03

The right reviewers

The procedure owner, site engineer and person responsible for the relevant integration.

Bring work your engineers already know.

Agree how quality and effort will be judged. You can book the conversation before sharing site documents.

Book a site walkthrough →

entangl.com/demo

SOURCE REGISTER

Incident reports 1-16

Reports and verification leads, not a census of distinct facility outages. Dates are reported onset dates; see the evidence labels and qualifications.

ID / onset	Provider / location	Reported condition	Evidence / sources
R01 2025-09-23	Equinix Metal New York	Top-of-rack switch instability.	Status Company/status / Archive
R02 2025-10-21	Equinix Metal Multiple metros	Backbone disruption; 15-minute interruption.	Status Company/status / Archive
R03 2025-10-23	Alaska Airlines Site undisclosed	Primary data center failure; U.S. site unconfirmed.	Geography lead Company/status / DCD / Coverage
R04 2025-11-28	CyrusOne / CME Aurora, IL	Cooling outage; roughly 10-hour trading halt.	Statement Company/status / Bloomberg / Coverage
R05 2025-12-02	Equinix Metal DC13	Backbone router instability.	Status Company/status / Archive
R06 2025-12-15	OVHcloud Vint Hill, VA	Rack V119G14 power fault; original not recovered.	Archive lead Archive
R07 2025-12-18	OVHcloud Vint Hill, VA	Rack V119G14 power fault; original not recovered.	Archive lead Archive
R08 2026-01-15	Equinix Metal Silicon Valley, CA	Metro connectivity loss; cause undisclosed.	Status Company/status / Archive
R09 2026-01-22	Equinix Metal DC10	Limited-host network instability.	Status Company/status / Archive
R10 2026-01-25	Oracle / TikTok U.S. site undisclosed	Weather-related power failure.	Statement Reuters/CNA / Coverage
R11 2026-01-28	Equinix Metal SV and DC metros	DDoS-related brief packet loss.	Archived status Archive
R12 2026-02-06	Equinix Metal Silicon Valley, CA	Network isolation; cause undisclosed.	Status Company/status / Archive
R13 2026-02-07	Azure West US, CA	Transformer/transfer failure and extended recovery.	Company RCA RCA / Video / DCD
R14 2026-03-03	Oracle / TikTok Ashburn, VA	Connectivity disruption; initiating cause unknown.	Status / statement Company/status / The Verge / Coverage
R15 2026-03-10	Nebius U.S. us-central1	Electrical maintenance and repeated interruptions.	Company RCA RCA
R16 2026-04-02	OVHcloud Hillsboro, OR	Rack IPMI access only; production outage unproven.	Management only Archive

R03: facility geography unconfirmed. R06-R07: original operator notices not recovered. R16: management access only. SV = Silicon Valley; DC = Washington, DC metro.

SOURCE REGISTER

Incident reports 17-32

Reports and verification leads, not a census of distinct facility outages. Dates are reported onset dates; see the evidence labels and qualifications.

ID / onset	Provider / location	Reported condition	Evidence / sources
R17 2026-04-14	Equinix Metal Chicago, IL	Device restart; packet loss on non-bonded hosts.	Status Company/status / Archive
R18 2026-05-07	AWS / Coinbase Northern Virginia	Data-hall cooling loss, per customer RCA.	Customer RCA RCA / DCD / Essay
R19 2026-05-29	Azure West US 2, WA	Voltage disturbances and cooling lockout.	Company RCA RCA / Video / Essay
R20 2026-06-30	Equinix Metal Atlanta, GA	AT4 isolated from the AT1 network point.	Status Company/status / Archive
R21 2026-07-23	Azure West US, CA	Optical repair automation removed egress routes.	Company RCA RCA / Video / DCD
R22 2026-08-03	CoreWeave US-EAST-04A	Power issue; may overlap R23.	Status Company/status / Archive
R23 2026-08-04	CoreWeave US-EAST-04A	Network hardware issue; relation to R22 unknown.	Status Company/status / Archive
R24 2026-08-13	RadiusDC / Namecheap Phoenix, AZ	Storm-related cooling loss; may overlap R25.	Company explanation Company/status / TechRadar
R25 2026-08-13	Blacksmith Phoenix, AZ	Cooling loss; operator unnamed; may overlap R24.	Company RCA RCA / Essay
R26 2026-08-15–16	OVHcloud Vint Hill, VA	Rack connectivity loss; archive date ambiguity.	Archived status Archive
R27 2026-08-17	Equinix Metal Silicon Valley, CA	All-customer-server impact; cause undisclosed.	Status Company/status / Archive
R28 2026-08-19	Nebius U.S. us-central1	Cooling loss and prolonged resource recovery.	Company RCA RCA
R29 2026-08-19	CoreWeave US-CENTRAL-08A	DH204: 19 GB300 racks lost capacity; cause unknown.	Archived status Company/status / Archive
R30 2026-08-20	Google Cloud Oregon	Fiber maintenance reduced inter-campus capacity.	Company RCA RCA
R31 2026-09-01	Google Cloud Iowa	Upgrade sequencing failure disconnected fibers.	Company RCA RCA / The Register
R32 2026-09-03	CoreWeave US-CENTRAL-08A	GPU fabric degraded; no total outage established.	Archived status Archive

Possible overlaps: R22-R23 and R24-R25. R29 onset precedes its status notice; notice duration is not outage duration. R32 describes degradation. R26 has archive date ambiguity.

READING THE ORIGINAL ACCOUNTS

The sources behind the case chapters.

Case	Primary account	Related material
01 / Nebius	August 19-20, 2026 Company post-mortem	Primary account linked
02 / Microsoft Azure	February 7-8, 2026 Company post-mortem	Company video Media coverage
03 / Microsoft Azure	May 29-30, 2026 Company post-mortem	Company video Industry essay
04 / Nebius	March 10-11, 2026 Company post-mortem	Primary account linked
05 / Microsoft Azure	July 23, 2026 Company post-mortem	Company video Media coverage
06 / Google Cloud	September 1, 2026 Company post-mortem	Media coverage
07 / Google Cloud	August 20, 2026 Company post-mortem	Primary account linked
08 / AWS / Coinbase	May 7-8, 2026 Customer post-mortem	Media coverage Industry essay
09 / Blacksmith	August 13-14, 2026 Company post-mortem	Industry essay

Additional perspectives

Reliability Engine: 2026 cooling-failure essay
Data Center Frontier: 2025 resilience review

The incident register includes media and archive links where located. A missing link means a suitable public document was not found in this research, not that no investigation exists. Some publisher pages may require a subscription.

Research cutoff: September 14, 2026. Durations follow the underlying account where status headers differ. Public source corrections and additional operator disclosures may change the record.

entangl

Put the lessons to work at your site.

One procedure.
One source set.
One result your engineers can assess.

43 **Review a maintenance procedure**

Preparation, review and the field handoff.

44 **Replay an incident**

The evidence available to the responder.

45 **Rehearse the recovery**

The dependencies between your teams.

Book your Entangl walkthrough →

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